

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2023 - 2024)

JEE(Advanced)**PRE-SCORE****20-11-2023****JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)**

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY**GENERAL :**

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

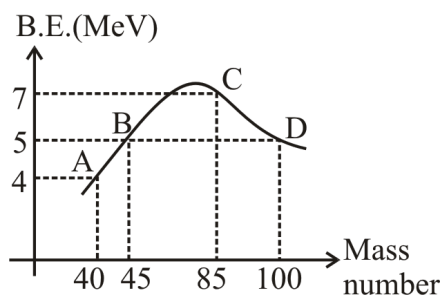
Negative Marks : -1 In all other cases

- The half-life of radioactive sample is 1620 years and its atomic mass is 226 The number of atoms that will decay from its 1 gm sample per second will be
 (A) 3.61×10^{10}
 (B) 3.6×10^{12}
 (C) 3.11×10^{15}
 (D) 31.1×10^{15}
- Kuldeep has a well and he constantly draws water from it with a volumetric flow rate q . Water is being constantly replenished into the well so the total volume stays constant at V . However, Kuldeep accidentally dropped a piece of radioactive bread into the well, which dissipated perfectly in the water. The bread contained radioactive isotopes with a half-life T . Kuldeep decided to ignore it and continued to draw water at the original rate. How long does it take until the radioactive activity of the well drops below A ? (N_0 are the initial no. of nuclei.)
 (A) $\frac{V}{q} \ln \left(\frac{N_0 \ln 2}{AT} \right)$
 (B) $\frac{1}{\frac{\ln 2}{T} - \frac{q}{V}} \ln \left(\frac{N_0 \ln 2}{AT} \right)$
 (C) $\frac{1}{\frac{\ln 2}{T} + \frac{q}{V}} \ln \left(\frac{N_0 \ln 2}{AT} \right)$
 (D) None

3. Far from a material with a thickness t , we place a source of gamma rays with an energy of approximately 2 MeV. The radiation has an absorption coefficient of μ_1 in the material. For radiation of this energy can happen that a photon converts into an electron-positron pair inside the material (the loss in intensity due to this effect is already included in the μ_1 coefficient). However, the positron immediately annihilates with an electron and produces 2 gamma flashes, each with energy $E = 511$ keV. The absorption coefficient of the material for the radiation with this energy is μ_2 . We place a detector far behind the material and observe the intensity of radiation of the energy E . For what thickness of material t will this intensity be the largest? Given that: $I = I_0 e^{-\mu x}$ where I_0 is the intensity before entering the material and x is the depth at which the intensity is calculated.

- (A) $\frac{3}{\mu_2 - \mu_1} \ln \left(\frac{\mu_2}{\mu_1} \right)$
 (B) $\frac{3}{\mu_2 - \mu_1} \ln \left(\frac{2\mu_2}{\mu_1} \right)$
 (C) $\frac{1}{\mu_2 - \mu_1} \ln \left(\frac{\mu_2}{\mu_1} \right)$
 (D) None

4. Binding energy per nucleon versus mass number curve for nuclei is shown in figure. A, B, C and D are four nuclei indicated on the curve. The Q value of following nuclear reaction $A + B \rightarrow C$ is $N \times 10^7$ eV. Find the value of N.



- (A) 210
 (B) 21
 (C) 2.1
 (D) None

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

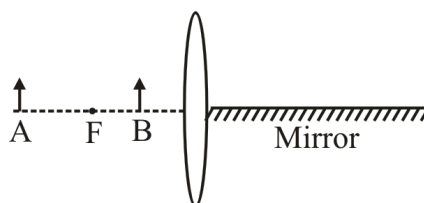
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

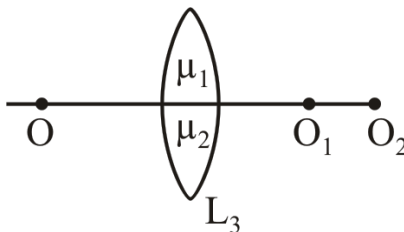
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Object-A is at a distance $2f$ from a convex lens of focal length f , and Object-B is at $f/2$ from the lens. On the other side of the lens there is a flat mirror laying parallel to and coincide with the optical axis of the lens. Consider refraction from lens and then reflection from plane mirror.



- (A) The images are formed at a distance of $2f$ and f from lens.
- (B) The magnitude of magnification is 1 in both cases
- (C) The magnitude of magnifications are 1 and 2.
- (D) Both the images are virtual.

6. We can imagine the electron (charge: e) as a thin spherical shell with uniformly distributed charge and radius R . For the electron, a magnetic moment can be created by making the sphere rotate around an axis passing through its centre. The electron's magnetic moment is known to be $\mu = \frac{eh}{4\pi m}$, where h is the Planck constant, (c : speed of light)
- (A) The maximum magnetic moment that the sphere can have is $\frac{1}{3}ecR$.
- (B) The minimum possible radius of the electron is $\frac{3h}{4\pi mc}$.
- (C) Maximum energy that can be stored in the electron is $\frac{ke^2}{4R}$.
- (D) none
7. Consider three converging lenses L_1 , L_2 and L_3 having identical geometrical construction. The index of refraction of L_1 and L_2 are μ_1 and μ_2 respectively. The upper half of the lens L_3 has a refractive index μ_1 and the lower half has μ_2 as shown in the figure. A point object O is imaged at O_1 by the lens L_1 and at O_2 by the lens L_2 placed in same position. If L_3 is placed at the same place,



- (A) There will be an image at O_1 .
- (B) There will be an image at O_2 .
- (C) The only image will form somewhere between O_1 and O_2 .
- (D) The only image will form away from O_2 .

SECTION-II (i) : (Maximum Marks: 12)

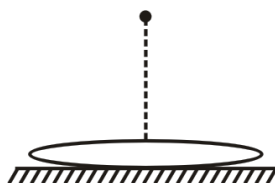
- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

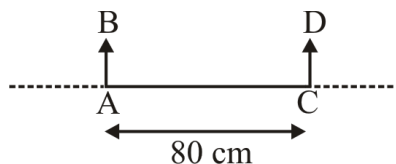
A thin equiconvex lens of refractive index $\frac{3}{2}$ is placed on a horizontal plane mirror as shown in the figure. The space between lens and the mirror is then filled with water of refractive index $\frac{4}{3}$. It is found that when a point object is placed 15 cm above the lens on the principal axis the object coincides with its own image.



1. At what distance xR object can be placed before water is filled so that image coincides with object if R is radius of curvature of each surface of lens. Find x .
2. What can be the value of R (in cm) from above information?

Paragraph for Questions 3 and 4

Two identical objects AB and CD each of height 1 mm are placed at separation of 80 cm. A converging lens of focal length of 30 cm with its principal axis along line AC is placed such that virtual image of point A and real image of point C coincide.



3. Distance (in cm) of point A from lens is :
4. If image of point B is B' and D is D' then length of B'D' (in mm) is :

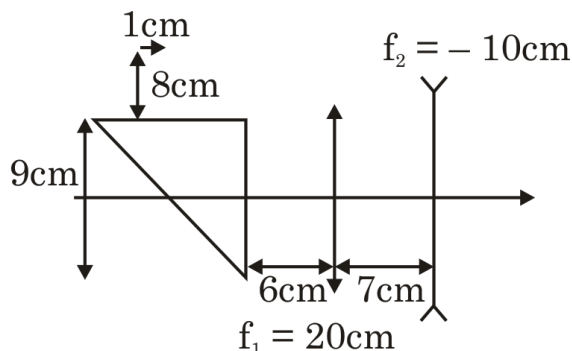
SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

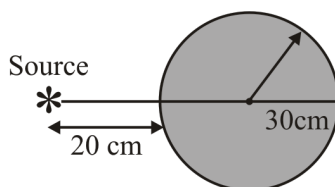
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. Consider two lenses at a mutual distance $d = 50$ cm. The first is a biconvex lens with focal length 20 cm, and the second is a biconcave lens with focal length -20 cm. Approximately how far (in cm) from the first lens do we need to place the object for the system to project it on itself?
6. A right-angled isosceles prism of side length 9cm and refractive index 1.5 is placed 6cm away from a converging lens of focal length $f_1 = 20$ cm, followed by a diverging lens of focal length $f_2 = -10$ cm a distance 7cm behind it. A 1cm stick is located 8cm above the prism, with one end aligned with the mid-point of the hypotenuse as shown in the figure below. The final image is formed at a distance x from the concave lens. Find x (in cm):



7. A point source of power 0.1W and emitting light of wavelength $5000 \text{ \AA}$ is placed at a distance 20 cm from a sphere of radius 30 cm as shown in the figure. The work function of the sphere is 1.5 eV. The efficiency of photoelectric emission of the sphere is 10^{-6} . If the time from the beginning of the experiment after which the photoelectric current stops is approximately $m \times 10^n$ sec in scientific notation. (m and n are integer). Find $|m| + |n|$



8. Chlorine molecule, which is moving with a velocity $v = 600$ m/s, absorbs a photon with a wavelength $\lambda = 350$ nm and is divided into two atoms. The first atom's velocity is measured to be $u = 1600$ m/s, which is perpendicular to the molecule's initial velocity. Find the chlorine molecule's binding energy if all the particles were at a state with minimal inner energy. It is presumed that a photon's momentum is negligible compared to chlorine's momentum. If Chlorine molecule's binding energy is approximately x eV then report $2x$ as answer.

9. For kinetic energy K_0 a free proton just is able to ionize a hydrogen atom. Assume that the electron inside the hydrogen atom is still and that the electromagnetic interaction between the atomic nucleus and the free proton is negligible. Hydrogen's binding energy is $E_0 = 13.6\text{eV}$, protons mass is $m_p = 1.67 \times 10^{-27}\text{ kg}$ and electrons mass $m_e = 9.11 \times 10^{-31}\text{ kg}$. If the smallest kinetic energy is $K_0\text{ keV}$ then report $4 K_0$ as answer. Round off to nearest integer.
10. A light ray starts from the interior of the circumference of a solid circle of radius R at $\pi/4$ radians with respect to the radial direction, radially inwards. Set the origin to be at the center of the circle. If the refractive index of the circle varies according to the relationship $n(r) = \sqrt{\frac{R^2}{r^2} + 1}$, determine the magnitude of the angular displacement (in rad) of the light ray when it reaches the center of the circle. (r is the distance from the centre of circle)

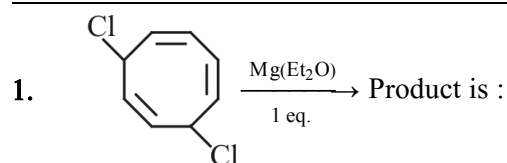
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

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- For each question, choose the option corresponding to the correct answer.
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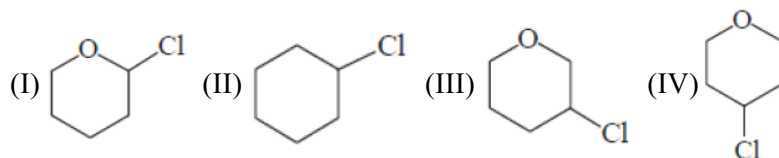
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases



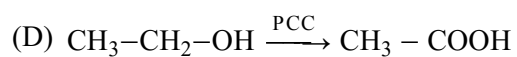
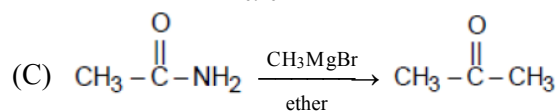
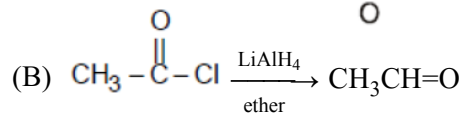
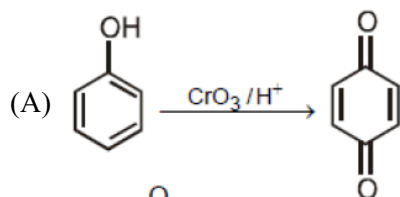
- (A) Anti-aromatic
(B) Non-aromatic
(C) Aromatic
(D) Quasi-aromatic

2. Correct reactivity order for S_N1 reaction

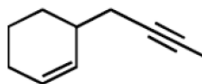


- (A) I > II > IV > III
(B) II > IV > III > I
(C) I > II > III > IV
(D) II > IV > I > III

3. Which reaction shows correct product.



4. The number of isomeric tetraenes (NOT containing sp-hybridized carbon atoms) that can be formed from the following reaction sequence is _____



(A) 2

(B) 4

(C) 3

(D) 1

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

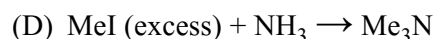
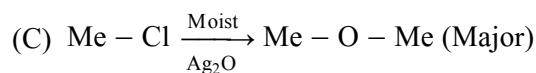
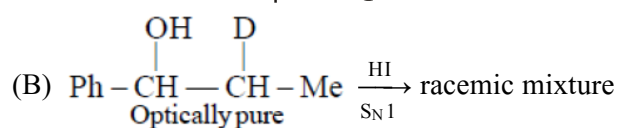
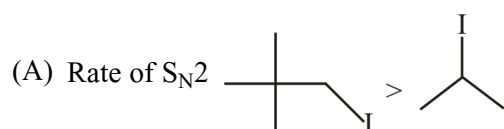
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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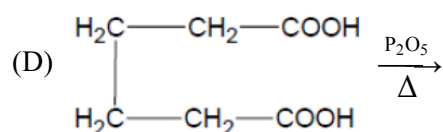
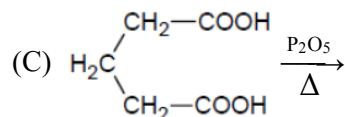
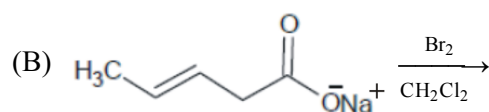
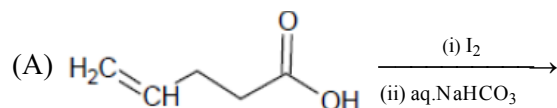
Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

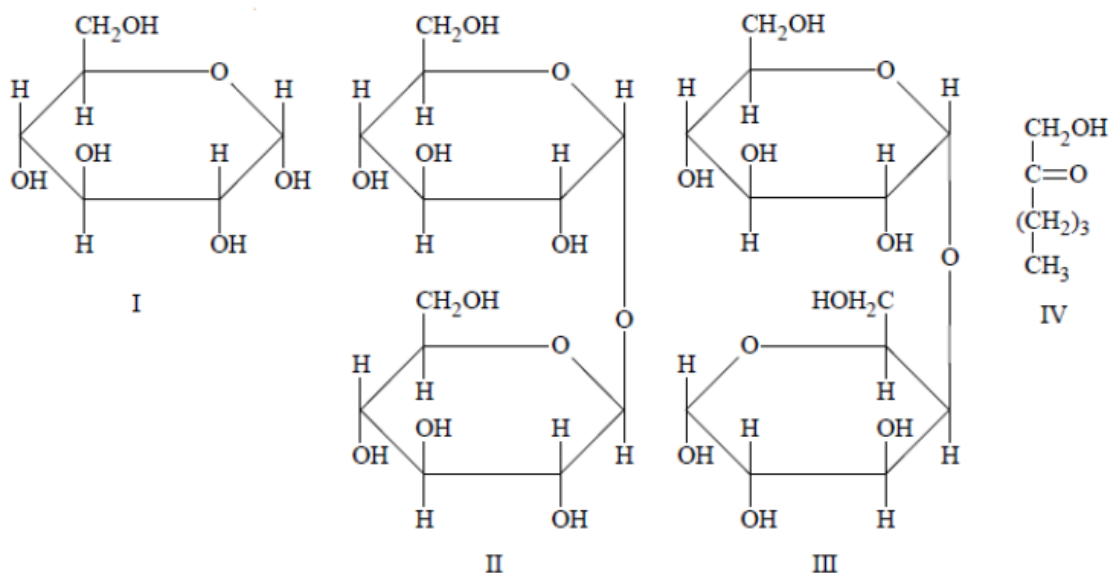
5. Choose the incorrect option(s)



6. Which of the following reaction(s) can generate a five membered ring as the major product?



7. Some organic compounds are given below



Select the correct statement about the above compounds from the following

- (A) Compounds I and III would reduce Fehling's solution and will also mutarotate.
 (B) Compound IV will not reduce Tollen's reagent and will not mutarotate.
 (C) Compound II will not be reducing but mutarotating.
 (D) The product(s) of the acid catalysed hydrolysis of both II and III will be reducing but not mutarotating.

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
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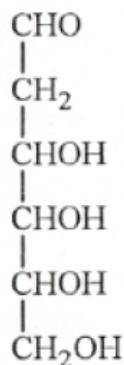
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Zero Marks : 0 In all other cases.

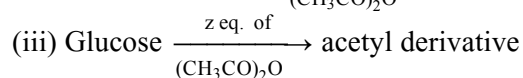
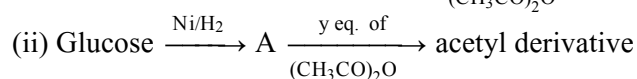
Paragraph for Questions 1 and 2

Carbohydrates are essential for life in both plants and animals. They form a major portion of our food. Honey has been used for a long time as an instant source of energy by 'Vaid' in ayurvedic system of medicine. Carbohydrates are used as storage molecules as starch in plants and glycogen in animals. Cell wall of bacteria and plants is made up of cellulose.

1. When the following aldohexose exists in its D-configuration, the total number of stereoisomers in its pyranose form is



2. Consider the following reactions :



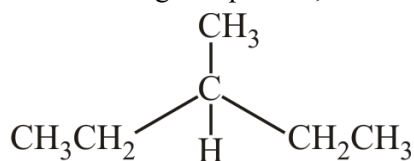
Report $x + y + z$?

Paragraph for Questions 3 and 4

In halogenation of an alkane, the alkane is said to undergo fluorination, chlorination, bromination or iodination depending on the identity of the halogen reactant. Chlorination and bromination are the two widely used alkane halogenation reactions. Fluorination reactions generally proceed too quickly to be useful and iodination reactions go too slowly.

Halogenations usually result in the formation of a mixture of products rather than a single product. More than one product results because more than one hydrogen atom on an alkane can be replaced with halogen atoms.

3. The maximum number of isomers (including stereoisomers) that are possible on -mono-chlorination of the following compounds, is :



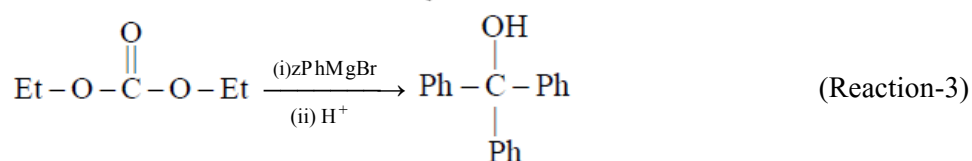
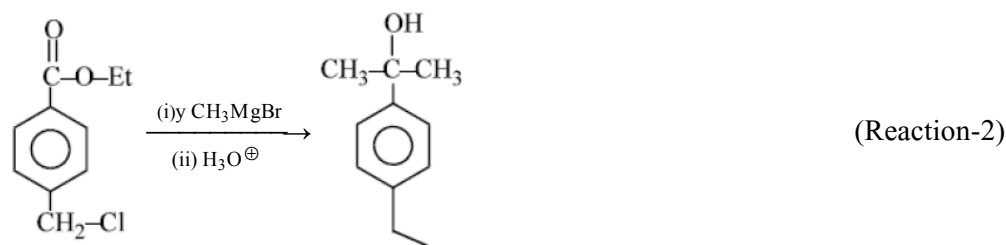
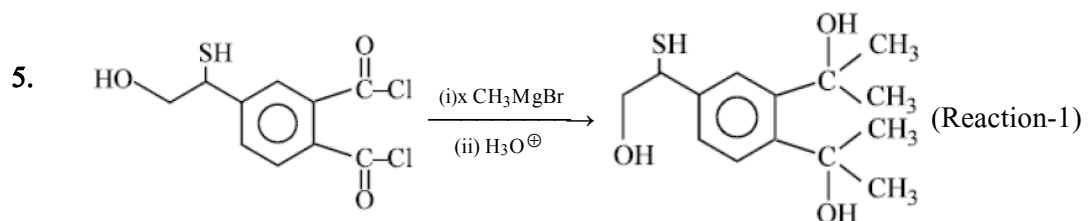
4. The maximum number of possible isomers (including stereoisomers) which may be formed on mono-bromination of 1-methylcyclohex-1-ene using NBS is _____

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

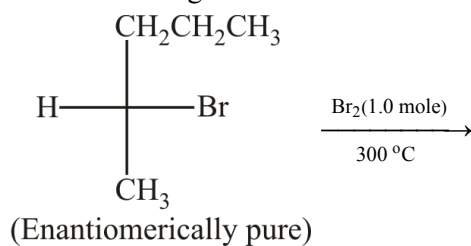
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases



What is the value of $x + y + z = ?$

6. In the following monobromination reaction, the number of possible chiral products is :



7. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is
8. How many organic compound(s) is/are correctly matched with number of chiral carbon(s) present in its structure :
- (i) Histamine = 0 (ii) Chloramphenicol = 2
(iii) Penicillin = 3 (iv) Aspartame = 4
(v) Sucralose = 9
9. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0% to the total weight of the hydrolysed products. The number of glycine units present in the decapeptide is
10. Degree of unsaturation in Aspirin is _____

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. If $[.]$ denotes the greatest integer function then the domain of the real valued function $f(x) = \log_{\left[x+\frac{1}{2}\right]} |x^2 - x - 2|$ is
- (A) $\left[\frac{3}{2}, \infty\right)$ (B) $\left[\frac{3}{2}, 2\right) \cup (2, \infty)$
(C) $\left(\frac{1}{2}, 2\right) \cup (2, \infty)$ (D) $(2, \infty)$
2. Number of solution of the equation $|\sin^{-1}(\sin x)| = \cos x$, for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is equal to
- (A) 0 (B) 2
(C) 4 (D) 6
3. Let a line ℓ through origin is tangent to the curve $y = x^3 + x + 16$. The slope of the line ℓ is
- (A) 2 (B) 8
(C) 13 (D) 26
4. If $\lim_{x \rightarrow 0} \frac{\tan^3 x + \tan^3 2x + \tan^3 3x}{\tan^3 4x + \tan^3 5x + \tan^3 6x} = \frac{p}{q}$ ($p, q \in \mathbb{N}$), then least value of $(p + q)$ is equal to
- (A) 45 (B) 47
(C) 49 (D) 51

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If $f(x) = \begin{cases} x \cdot \sin\left(\frac{\pi}{x}\right) & , \quad x \neq 0 \\ k & , \quad x = 0 \end{cases}$ then

- (A) for $k = 1$, $f(x)$ is discontinuous at $x = 0$
- (B) for $k = 1$, $f(x)$ is continuous at $x = 0$
- (C) for $k = \pi$, $f(x)$ is continuous at $x = 0$
- (D) for $k = 0$, $f(x)$ is continuous at $x = 0$

6. For a certain curve $y = f(x)$ satisfying $\frac{d^2y}{dx^2} = 6x - 4$, $f(x)$ has local minimum value 5 when $x = 1$ then which of the following is/are correct ?
- (A) Equation of curve y is $y = x^3 - 2x^2 + x + 5$
- (B) $f(x)$ has a local maximum at $x = \frac{1}{3}$
- (C) Equation of curve is $y = x^3 - 2x^2 - x + 7$
- (D) The number of real and distinct roots of $f(x) = 0$ is 1
7. The value of 'x' satisfying $\sin(2 \cos^{-1}(\cot(\tan^{-1}x))) = 0$ is/are
- (A) 1
- (B) 0
- (C) -1
- (D) $\frac{1}{2}$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider the function $f(x) = x|x^2 - x| + |x - 2|$.

1. The total number of local maxima and local minima of $f(x)$ is
2. If $x = \alpha$ is the point of maxima of the function $f(x)$, then $|6\alpha| =$

Paragraph for Questions 3 and 4

Let $f(x) = x^2 e^x - 2e^x + 3$, $x \in (-4, 1)$

3. Number of real roots of the equation $f(x) = 0$ is
4. Number of local maxima of the function $f(x)$ is

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

5. 'p' is number of points where $y = (x^3 + 1)|x^2 - 1| + \sin|x + 1|$ is not differentiable 'q' is number of points where $y = \operatorname{sgn}(x^3 + x^2 - x + 3)$ is discontinuous and $r = \sqrt{3} \tan \left(\cos^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{-1}{\sqrt{3}} \right) \right)$ then $p + q + r$ is equal to

6.
$$f(x) = \begin{cases} 5 & x = 0 \\ x^2 + 5x + \alpha & 0 < x < 2 \\ \beta x + \gamma & 2 \leq x \leq 3 \end{cases}$$

If $f(x)$ satisfies conditions of LMVT in $[0, 3]$ then $2\alpha - \beta + \gamma + f'(1) + \frac{f'(3)}{9}$ is :

7. $\lim_{x \rightarrow 0} \left(\frac{1 - \cos x \sqrt{\cos 2x}}{x^2} + \sqrt[3]{\frac{1 + \cos^2 x + 3\cos^4 x + 3\cos x}{\cos x + 63}} \right)$ is equal to

8. Let $f(x) = \tan^{-1}(\cot x - 2\cot 2x)$ and $\sum_{r=1}^5 f(r) = a - b\pi$, where $a, b \in \mathbb{N}$. Find the value of $(a + b)$

9. For any real number x , let $[x]$ denote the largest integer less than or equal to x . Let f be a real valued function defined on the interval $[-3, 3]$ by $f(x) = \begin{cases} -x - [-x] & \text{if } [x] \text{ is even} \\ x - [x] & \text{if } [x] \text{ is odd} \end{cases}$.

If L denotes the number of points of discontinuity and M denotes the number of points of non-derivability of $f(x)$, then find $(L + M)$.

10. If p_1 and p_2 are the length of the perpendiculars from origin on the tangent and normal drawn to the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 6^{\frac{2}{3}}$, then find the value of $\frac{(4p_1^2 + p_2^2)}{6}$.

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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E-24/24

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